

Bayesian Synthesis for Phylogenetic Diversification Rate Models

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Abstract

We present a Bayesian program synthesis framework for automatically discovering phylogenetic diversification rate models from observed tree data. Models are represented as expressions in a probabilistic domain-specific language (DSL) generated by a context-free grammar (CFG). Synthesis is performed via Markov chain Monte Carlo (MCMC) over the space of DSL expressions.

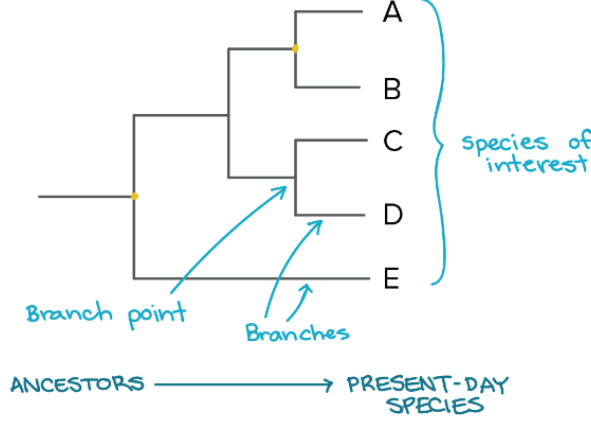
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1 Method Overview

1.1 Problem Statement

Given a reconstructed phylogenetic tree consisting of a topology and branch lengths for n extant species, the goal is to automatically infer both the *form* (structure) and the *parameters* of the diversification rate model that best explains the observed tree under a birth–death process. Here is a picture of the tree data we are modeling:



1.2 Domain-Specific Language

We define a probabilistic DSL whose expressions are generated by the following context-free grammar G :

$$\begin{aligned}
 N_1 &::= (\text{NoisyPhylo } N_2 \ N_3) \\
 N_2 &::= (\text{Noise } \square) \\
 N_3 &::= (\text{CRB } \square) \mid (\text{CRBD } \square \ \square) \mid (\text{TDB } \square \ \square) \mid (\text{TDBD } \square \ \square \ \square) \mid (\text{BAMM } \square \ N_3 \ N_3)
 \end{aligned} \tag{1}$$

where $\square$ denotes parameter holes filled by draws from continuous prior distributions, and N_1 is the start symbol. Each production of N_3 corresponds to a diversification model class:

- CRB: constant-rate birth (Yule) process;
- CRBD: constant-rate birth–death;
- TDB: time-dependent birth (no extinction);
- TDBD: time-dependent birth–death;
- BAMM: introduces a Poisson change-point switching between a background (bg) and foreground (fg) sub-model, each of which is itself an N_3 expression (allowing recursive nesting).

1.3 Prior Distribution

Each production rule r_{ik} for non-terminal N_i is assigned a selection probability $p_{ik} > 0$. Parameters are drawn independently from specified prior distributions (exponential, normal, beta, etc.). The prior probability of an expression $E = (s, \theta)$ with structure s and parameters θ factorizes as $\text{PRIOR}[E] = p_s \cdot \gamma_s(\theta)$, where p_s is the product of selection probabilities along the derivation and γ_s is the product of parameter densities.

1.4 Likelihood

Each expression E defines a generative process over phylogenetic trees.

The likelihood $\text{LIKELIHOOD}[E](O)$ of an observed tree O is computed via:

- Closed-form expressions for pure-birth models.

- A unified ODE framework for models involving extinction and/or regime switching.
- A contamination-mixture noise model that interpolates between a structured model likelihood and a baseline noise likelihood.

Remark. It is important to note that computing the likelihood for an isolated segment or a partial subtree is ill-posed. Because reconstructed phylogenetic trees are retrospectively conditioned on the sampling of extant species at the present day, the likelihood computation is only well-defined and meaningful when evaluated globally over the entire tree.

2 Convergence Theory

This section establishes that the MCMC synthesis algorithm converges to the posterior distribution over DSL expressions. We follow the general framework of your PhD dissertation on Bayesian synthesis [1], adapting the formalization to the phylogenetic setting.

2.1 Probabilistic DSL Formalization

Definition 2.1 (Phylogenetic DSL). *The phylogenetic domain-specific data modeling language D is comprised of:*

- A countable structure space $S = \mathcal{L}(G)$, the set of all s -expressions with parameter holes derivable from G ;
- For each $s \in S$, a parameter space Θ_s equipped with sigma-algebra $\mathcal{F}_s$ and base measure λ_s (products of Lebesgue measures on $\mathbb{R}$);
- An observation space $\mathcal{X}$ consisting of all valid reconstructed phylogenetic trees with extant tips (Definition 2.2).

Definition 2.2 (Observation space). *An observation $O \in \mathcal{X}$ is a rooted phylogenetic tree with $n \geq 2$ extant tips. Concretely, O is characterized by:*

- A rooted labeled bifurcating topology $\mathfrak{t}$ with n leaves and $n - 1$ internal nodes $V_{\text{int}} = \{v_1, \dots, v_{n-1}\}$;
- Internal-node ages $(\tau_{v_1}, \dots, \tau_{v_{n-1}}) \in \mathbb{R}_{>0}^{n-1}$ satisfying $\tau_{\text{pa}(v)} > \tau_v$ for every non-root $v \in V_{\text{int}}$ (all leaves have age 0).

Each branch length is determined by $\ell_e = \tau_{\text{pa}(e)} - \tau_{\text{child}(e)}$. The tree height is the root age $T = \max_v \tau_v$, the total branch length is $S = \sum_e \ell_e$, and the branching times are the sorted internal-node ages $\tau_1 > \tau_2 > \dots > \tau_{n-1} > 0$.

Definition 2.3 (Likelihood semantics). *For each expression $E \in \mathcal{L}$ and observation $O \in \mathcal{X}$, the likelihood $\text{LIKELIHOOD}[E](O) \geq 0$ evaluates how well the diversification model specified by E explains the observed tree O .*

2.2 Conditions for Well-Defined Posterior

Three conditions on the PRIOR and LIKELIHOOD semantics ensure that the posterior $\text{POST}[E](O)$ is a well-defined probability distribution for any fixed observation O .

Condition 2.4 (Normalized Prior). *The prior induces a probability distribution over $(\mathcal{L}, \Sigma_{\mathcal{L}})$:*

$$\sum_{s \in S} \int_{\theta \in \Theta_s} \text{PRIOR}[(s, \theta)] \lambda_s(d\theta) = 1. \quad (2)$$

Proposition 2.5 (Grammar Consistency). *The grammar G in (1), is consistent in the sense of Booth and Thompson.*

Proof. The non-terminals N_1 and N_2 are non-recursive (each has a single production with no non-terminal children), so it suffices to verify consistency of the N_3 sub-grammar.

$$\begin{array}{lcl}
N_3 \rightarrow H_1 H_0 & \frac{5}{20} \\
| H_2 H_0 & \frac{5}{20} \\
| H_3 H_0 & \frac{4}{20} \\
| H_4 H_0 & \frac{4}{20} \\
| H_5 N_3 & \frac{2}{20}
\end{array}$$

$$\begin{array}{ll}
H_0 \rightarrow \square & H_1 \rightarrow \text{CRB} \\
H_2 \rightarrow H_6 H_0 & H_6 \rightarrow \text{CRBD} \\
H_3 \rightarrow H_7 H_0 & H_7 \rightarrow \text{TDB} \\
H_4 \rightarrow H_8 H_0 & H_8 \rightarrow H_{10} H_0 \\
H_5 \rightarrow H_9 N_3 & H_9 \rightarrow H_{11} H_0 \\
H_{10} \rightarrow H_{12} H_0 & H_{11} \rightarrow \text{BAMM} \\
H_{12} \rightarrow \text{TDBD}
\end{array}$$

Following Gecse and Kovács, define $m_{ij} = \sum_{k=1}^{r_i} p_{ik} n_{ikj}$, The expectation matrix $M = [m_{ij}]$ is:

$$M = \begin{pmatrix} \frac{2}{20} & \frac{18}{20} & \frac{5}{20} & \frac{5}{20} & \frac{4}{20} & \frac{4}{20} & \frac{2}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3)$$

The nonzero eigenvalues of M are 0.3702... and -0.2702... which lie in the interval $[-1,1]$. By Theorem 3.35 of [1], G is consistent. $\square$

Since G is consistent and the parameter distributions μ_{ik} are probability measures, Condition 2.4 holds by the factorization $\text{PRIOR}[(s, \theta)] = p_s \cdot \gamma_s(\theta)$ and the fact that $\sum_{s \in S} p_s = 1$ and $\int_{\Theta_s} \gamma_s(\theta) \lambda_s(d\theta) = 1$.

Condition 2.6 (Positive Likelihood). *For each observation $O \in \mathcal{X}$, there exists a set $A \subset \mathcal{L}$ of positive prior measure such that $\text{LIKELIHOOD}[E](O) > 0$ for all $E \in A$.*

Proof. Since $S > 0$ for any valid tree, the CRB likelihood $\Gamma(n) \lambda^{n-1} e^{-\lambda S}$ is strictly positive for all $\lambda > 0$. And the set of CRB expressions has positive prior measure. $\square$

Condition 2.7 (Bounded Likelihood). *For each observation $O \in \mathcal{X}$, the likelihood is essentially bounded: there exists a finite constant $c_O^{\max} > 0$ such that $\{E \in \mathcal{L} \mid \text{LIKELIHOOD}[E](O) > c_O^{\max}\}$ has prior measure zero.*

Proof. Let $E = (\text{NoisyPhylo}(\text{Noise } \sigma) M)$ with $\pi = \sigma/(1 + \sigma) \in [0, 1)$. The final likelihood satisfies

$$\text{LIKELIHOOD}[E](O) = (1 - \pi) \mathcal{L}_{\text{model}}(M, O) + \pi \mathcal{L}_{\text{noise}}(O) \leq \mathcal{L}_{\text{model}}(M, O) + \mathcal{L}_{\text{noise}}(O). \quad (4)$$

The noise baseline is $\mathcal{L}_{\text{noise}}(O) = \beta [\Gamma(n)]^2 / (S + \beta)^n$, where $\beta = 1/\text{scale}_\lambda$ is a fixed hyperparameter of the prior, and n, S are determined by the fixed tree O . For any given O , this is a finite positive constant independent of the expression E .

It remains to show that $\mathcal{L}_{\text{model}}(M, O)$ is bounded independently of the expression M . We treat each model class in turn.

Pure-birth models (CRB, TDB). Both have closed-form likelihoods of the form $\mathcal{L} = \Gamma(n) \prod_{k=1}^{n-1} \lambda(\tau_k) \cdot e^{-\sum_i \Lambda_i}$, where $\Lambda_i = \int_{\tau_c^{(i)}}^{\tau_p^{(i)}} \lambda(\tau) d\tau > 0$. For CRB ($\lambda(\tau) \equiv \lambda$), this gives $\Gamma(n) \lambda^{n-1} e^{-\lambda S}$, maximized at $\lambda^* = (n-1)/S$ with finite maximum $\Gamma(n) ((n-1)/(eS))^{n-1}$, independent of λ . For TDB ($\lambda(\tau) = \lambda_0 e^{z(T-\tau)}$), the likelihood is a continuous function that vanishes at every boundary of the parameter space, so it attains a finite supremum.

Birth-death models (CRBD, TDBD, BAMM). The likelihood takes the ODE-based form $\Gamma(n) D_{\text{root}} / (1 - E(T))^2$. The ODE solutions D_{root} and $E(T)$ are continuous functions of the rate parameters. At degenerate boundaries ($\lambda \rightarrow 0$, $\mu \rightarrow \infty$, or $E(T) \rightarrow 1$) the likelihood vanishes ($\log \mathcal{L} = -\infty$); at the opposite extreme ($\mu = 0$) it reduces to a pure-birth model bounded as above. So they attain finite suprema. $\square$

Proposition 2.8 (Well-Defined Posterior). *If Conditions 2.4, 2.6 and 2.7 hold and the likelihood is measurable, then for any fixed observation $O \in \mathcal{X}$, the posterior $\text{POST}[E](O) \propto \text{LIKELIHOOD}[E](O) \cdot \text{PRIOR}[E]$ is a well-defined probability distribution over $(\mathcal{L}, \Sigma_{\mathcal{L}})$.*

Proof. The proof is identical to Propositions 3.11–3.12 of [1]: Condition 2.6 gives $c_O > 0$ and Condition 2.7 with Condition 2.4 gives $c_O < \infty$. $\square$

2.3 Transition Operator

Our transition operator is an instance of Algorithm 3.3 of [1] applied to the phylogenetic grammar G : uniformly select a node a in the parse tree, sever the subexpression at a to obtain a non-terminal N_i , re-expand from N_i via EXPAND, and accept the proposal with Metropolis–Hastings probability

$$\alpha(E, E') = \min \left\{ 1, \frac{|A_E|}{|A_{E'}|} \cdot \frac{\text{LIKELIHOOD}[E'](O)}{\text{LIKELIHOOD}[E](O)} \right\}, \quad (5)$$

where $|A_E|$ is the number of nodes in the parse tree of E .

Since our grammar is context-free, the three conditions required by the MCMC convergence theory are all satisfied by the proofs of Lemmas 3.50–3.52 of [1], which apply to any context-free probabilistic DSL without modification.

2.4 Main Convergence Results

Theorem 2.9 (MCMC Convergence in Total Variation). *Let $O \in \mathcal{X}$ be an observation and $E_0 \in \mathcal{L}$ satisfy $\text{LIKELIHOOD}[E_0](O) > 0$. Let $(E_1, \dots, E_n)$ be the output of the MCMC algorithm (Algorithm 3.1 of [1]) using the transition operator described above. Then*

$$\sup_{E \in \mathcal{L}} \left| \sum_{E \in \mathcal{E}} [\text{ApproxPost}_{E_0}^{(n)}[E](O) - \text{POST}[E](O)] \right| \xrightarrow{n \rightarrow \infty} 0. \quad (6)$$

Proof. Lemmas 3.50–3.52 of [1] verify posterior invariance, irreducibility, and aperiodicity for the transition operator. The result follows from Theorem 3.19 of [1]. $\square$

Theorem 2.10 (Strong Law of Large Numbers for MCMC). *Under the hypotheses of Theorem 2.9, for any function $\varphi : \mathcal{L} \rightarrow \mathbb{R}$ satisfying $\sum_{E \in \mathcal{L}} |\varphi(E)| \cdot \text{POST}[E](O) < \infty$,*

$$\frac{1}{n} \sum_{i=1}^n \varphi(E_i) \xrightarrow{n \rightarrow \infty} \sum_{E \in \mathcal{L}} \varphi(E) \cdot \text{POST}[E](O) =: \mathbb{E}_{\text{POST}}[\varphi(E)] \quad \text{almost surely.} \quad (7)$$

Proof. Follows from Theorem 3.20 of [1]. □

3 Evaluation & Questions

To verify the correctness of the algorithm, we generate a phylogenetic tree from a known model structure and run the MCMC algorithm to attempt to recover this true structure.

The original structure is CRBD(0.09, 0.02), noise=0, log L = -12.0640.

But the best expression found is (BAMM 0.0311 (TDBD 0.1462 -0.4111 0.0783) (CRBD 0.1164 0.0073)), noise=0.0207, log L = -8.9632

This raises several questions:

- **Maximum Likelihood?** Why does the global maximum-likelihood expression differs from the true generative structure? Does this discrepancy indicate a bug in our implementation, or a flaw in the convergence proofs established in Section 2? Or is this behavior entirely expected?
- **Overfitting?** Do complex structures naturally achieve higher likelihoods simply by having more free parameters to fit stochastic noise? Our attempt runs show that data generated by CRBD correctly yields CRBD as the most frequently visited structure on the chain, but with less log-likelihood than BAMM. Does this imply we should adopt the highest posterior frequency structure as our primary result, while maintaining a separate maximum-likelihood expression for each expression class?

Acknowledgments

The framework closely follows methods introduced in [1].

We are very impressed by your dissertation on Bayesian synthesis.

References

- [1] Feras Ahmad Khaled Saad, *Scalable Structure Learning, Inference, and Analysis with Probabilistic Programs*, Ph.D. thesis, MIT, 2022. Chapter 3: Synthesizing Probabilistic Programs in Domain-Specific Modeling Languages.